Computer Networks Laboratory

Lab Report #4

Assigning IP Addresses to PCs

Course:	Computer Networks
Tool:	Cisco Packet Tracer
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Level:	Undergraduate

Lab Summary

This lab covers three methods of assigning IP addresses to end devices:
Static CLI, **Static GUI**, and **Dynamic (DHCP)**.
Each method is demonstrated and verified using Cisco Packet Tracer.

Contents

1. Introduction

In any IP-based network, every end device must be assigned a unique IP address to communicate with other nodes. Without proper addressing, packets cannot be routed, hosts cannot be identified, and network services cannot function. This laboratory explores the fundamental mechanisms available for assigning IP addresses to end devices in a simulated Cisco network environment.

Three distinct methods are examined:

- **Static IP Assignment via CLI** — manually entering the IP configuration using the command-line interface of the simulated PC.
- **Static IP Assignment via GUI** — using the graphical Desktop interface in Cisco Packet Tracer to set IP parameters.
- **Dynamic IP Assignment via DHCP** — leveraging a DHCP server (configured in Lab 5) to automatically distribute IP addresses to clients.

Understanding these methods is foundational to any networking course, as address management is a core skill for network administrators.

2. Objective

The primary objectives of this lab are:

1. Learn how to manually configure an IP address, subnet mask, and default gateway on a PC using the command-line interface (CLI).
2. Learn how to configure the same parameters through the graphical user interface (GUI) in Cisco Packet Tracer.
3. Understand the role of DHCP in dynamic address allocation and configure a PC to obtain an address automatically.
4. Verify the applied configuration using the `ipconfig /all` command and interpret its output.

3. Network Topology

The lab uses a simple LAN topology consisting of the following devices:

Table 1: Network Devices Used in Lab 4

Device	Type	Interface	Role
Router3	Cisco 2911 Router	Gig0/0, Gig0/1	Default Gateway / DHCP Server
2960-24TT	Cisco Catalyst Switch	Fa0/1 – Fa0/4	Layer 2 Switching
PC2	PC-PT End Device	FastEthernet0	Network Client
PC3	PC-PT End Device	FastEthernet0	Network Client
PC4	PC-PT End Device	FastEthernet0	Network Client

Table 2: IP Addressing Plan

Device	IP Address	Subnet Mask	Default Gateway
Router3 (Gig0/0)	192.168.1.1	255.255.255.0	—
PC2	192.168.1.10	255.255.255.0	192.168.1.1
PC3	192.168.1.11	255.255.255.0	192.168.1.1
PC4	192.168.1.12	255.255.255.0	192.168.1.1

4. Background & Theory

4.1. IP Addressing Fundamentals

An IPv4 address is a 32-bit numeric label written in dotted-decimal notation (e.g., `192.168.1.10`). It uniquely identifies a host on a network. Three parameters are required for basic IP connectivity:

- **IP Address** — uniquely identifies the device on the network.
- **Subnet Mask** — defines which portion of the address is the network identifier and which is the host identifier.
- **Default Gateway** — the IP address of the router interface that forwards packets destined for external networks.

4.2. Static vs. Dynamic Assignment

Table 3: Comparison of IP Assignment Methods

Criterion	Static	Dynamic (DHCP)
Configuration	Manual entry per device	Automatic via server
Scalability	Poor for large networks	Excellent
IP Conflicts	Possible if not managed	Avoided by server
Use Case	Servers, printers, routers	End-user PCs, laptops
Reliability	Always available	Depends on DHCP server

4.3. DHCP Protocol Overview

DHCP (Dynamic Host Configuration Protocol, RFC 2131) automates the assignment of IP configuration to clients. The exchange follows the **DORA** process:

1. **Discover** — Client broadcasts a request for an available DHCP server.
2. **Offer** — Server proposes an IP address and configuration parameters.
3. **Request** — Client formally requests the offered address.
4. **Acknowledge** — Server confirms the lease and sends the full configuration.

5. Configuration Steps

5.1. Method 1 — Static IP Assignment via CLI

In Cisco Packet Tracer, each PC has a built-in command prompt (Desktop → Command Prompt). The `ipconfig` command is used with positional arguments to set the IP address, subnet mask, and default gateway simultaneously.

```
PC> ipconfig 192.168.1.10 255.255.255.0 192.168.1.1
```

Listing 1: Static IP Configuration via CLI (PC2)

Parameter breakdown:

Parameter	Description
`192.168.1.10`	IPv4 address assigned to the PC
`255.255.255.0`	Subnet mask (Class C, /24 prefix)
`192.168.1.1`	Default gateway (router interface IP)

Note: The Packet Tracer `ipconfig` CLI syntax differs from real Windows `netsh` or Linux `ip addr` commands. In a real Windows environment, static IP is configured via `netsh interface ip set address` or through the Network Adapter settings.

5.2. Method 2 — Static IP Assignment via GUI

Cisco Packet Tracer provides a graphical IP Configuration panel accessible from each PC's desktop.

Steps:

1. Click on the PC device to open its configuration window.
2. Navigate to **Desktop** → **IP Configuration**.
3. Select the **Static** radio button.
4. Enter the following values in the respective fields:

Table 4: GUI IP Configuration Parameters

Field	`Value`
IP Address	`192.168.1.10`
Subnet Mask	`255.255.255.0`
Default Gateway	`192.168.1.1`
DNS Server	`8.8.8.8`

Both CLI and GUI methods produce the same result. The GUI method is generally preferred by beginners due to its intuitive interface, while the CLI method is faster for experienced administrators and scriptable in real environments.

5.3. Method 3 — Dynamic IP Assignment via DHCP

After the DHCP server has been configured on Router3 (as performed in Lab 5), end devices can be set to obtain their IP configuration automatically.

5.3.1. Enable DHCP on a PC via CLI

```
PC> ipconfig /dhcp
```

Listing 2: Enable DHCP on a PC (Command Prompt)

This command instructs the PC to release any current static configuration and send a DHCP Discover broadcast. If a DHCP server is reachable, the PC will receive an IP address lease automatically.

5.3.2. Enable DHCP on a PC via GUI

1. Open the PC → **Desktop** → **IP Configuration**.
2. Select the **DHCP** radio button.
3. The fields will populate automatically once the DHCP server responds.

5.3.3. Router DHCP Server Configuration (Lab 5 Reference)

For completeness, the commands used on Router3 to create the DHCP pool are shown below:

```
Router> enable
Router# configure terminal

! Exclude the router's own IP and reserved addresses
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.9

! Create the DHCP pool
Router(config)# ip dhcp pool LAN_POOL
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# lease 7
Router(dhcp-config)# exit

! Configure the router interface facing the LAN
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# end
Router# write memory
```

Listing 3: DHCP Server Configuration on Router3

6. Verification

After applying any of the three methods, the configuration must be verified. The `ipconfig /all` command displays the full IP configuration of all network interfaces on the PC.

6.1. Verification Command

```
PC> ipconfig /all
```

Listing 4: Verification using ipconfig /all

6.2. Expected Output

```
FastEthernet0 Connection:

   Connection-specific DNS Suffix  . :
   Physical Address. . . . . . . . . : 00D0.9729.XXXX
   Link-local IPv6 Address . . . . . : FE80::XX
   IP Address. . . . . . . . . . . . : 192.168.1.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
   DNS Servers . . . . . . . . . . . : 0.0.0.0
   DHCP Enabled. . . . . . . . . . . : No
   DHCP Server . . . . . . . . . . . : 0.0.0.0
```

Listing 5: Expected ipconfig /all Output (Static Configuration)

Output for DHCP-assigned configuration:

```
FastEthernet0 Connection:

   IP Address. . . . . . . . . . . . : 192.168.1.11
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Enabled. . . . . . . . . . . : Yes
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   Lease Obtained  . . . . . . . . . : [Timestamp]
   Lease Expires . . . . . . . . . . : [Timestamp + 7 days]
```

Listing 6: Expected ipconfig /all Output (DHCP Configuration)

6.3. Field Interpretation

Table 5: Output Field Descriptions

`Field`	Meaning
`IP Address`	The IPv4 address currently assigned to the interface
`Subnet Mask`	Defines the network and host portions of the address
`Default Gateway`	Router IP used for traffic destined outside the local subnet
`DHCP Enabled`	`Yes` if address obtained via DHCP; `No` if static
`DHCP Server`	IP of the DHCP server that issued the lease
`Physical Address`	MAC address of the network interface card

6.4. Connectivity Test

After verifying the IP configuration, test end-to-end connectivity using the `ping` command:

```
! Ping the default gateway
PC> ping 192.168.1.1

! Ping another PC on the same subnet
PC> ping 192.168.1.11
```

Listing 7: Connectivity Test between PCs

Expected Result: A successful ping shows `Reply from 192.168.1.X` with 0% packet loss, confirming that addressing and routing are correctly configured.

7. Discussion

7.1. Advantages and Disadvantages of Each Method

Static CLI: The command-line method is fast and reproducible, making it suitable for automation and bulk configuration scripts. However, it requires precise syntax and offers no validation feedback.

Static GUI: The graphical panel is more intuitive for beginners and reduces typographical errors. The trade-off is that it does not scale well in large networks, as each device must be configured individually.

DHCP: Dynamic assignment drastically reduces administrative overhead in networks with many hosts. Centralized control from the DHCP server ensures there are no IP conflicts and simplifies moves, additions, and changes. The primary dependency is the availability of the DHCP server — if it is unreachable, clients may fail to obtain an address.

7.2. Security Considerations

- **DHCP Snooping** should be enabled on the switch to prevent rogue DHCP servers from distributing malicious configurations.
- **Static IP** configurations are more predictable and can be combined with port security on switches for enhanced control.
- Always document static IP assignments in a network addressing table (IPAM) to prevent conflicts.

8. Troubleshooting

Table 6: Common Issues and Solutions

Symptom	Likely Cause	Solution
Ping fails to gateway	Wrong default gateway	Verify gateway IP matches router interface
DHCP address not obtained	DHCP server off / misconfigured	Check router DHCP pool and excluded range
IP conflict detected	Duplicate static IPs	Review addressing table; use DHCP exclusions
169.254.x.x address shown	APIPA fallback (no DHCP response)	Verify DHCP server is reachable
Interface shows `down`	Cable disconnected in simulation	Reconnect cable in Packet Tracer topology

9. Conclusion

This lab demonstrated three methods of assigning IP addresses to end devices in a simulated Cisco network:

1. **Static CLI** — efficient and scriptable but requires careful manual input.
2. **Static GUI** — user-friendly but limited in scalability.
3. **DHCP** — the preferred method for dynamic, large-scale environments, relying on a centralized server to manage address allocation automatically.

The verification procedure using `ipconfig /all` confirmed that all three methods produce a correctly configured network interface, enabling full IP communication within the LAN. Mastery of these techniques is essential for any network administrator, as IP address management is a daily operational task in real-world networking.

References

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4. Odom, W. *CCNA 200-301 Official Cert Guide*, Volume 1. Cisco Press, 2019.